

Replay-Aware TSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: tsplib_failure_replay.
- Best TSPLIB holdout gap: tsplib_failure_replay.
- Best synthetic holdout gap: tsplib_failure_replay_compression.

Run Metadata

- run_name: run_20260513_140147_i
- started_at_local: 2026-05-13 14:01:47
- finished_at_local: 2026-05-13 14:11:58
- duration_hhmm: 00:10
- duration_seconds: 611.317
- seed_offset: 8000
- replicate_label: i
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
tsplib_no_replay	none	score_only	False	0.213387	0.064916	0.159398	0.778745	0.56	-0.031981
tsplib_random_replay	random	score_only	False	0.213387	0.064916	0.159398	0.822825	0.56	-0.030268
tsplib_failure_replay	failure	score_only	False	0.087678	0.010343	0.059556	0.80747	0.76	0.058506
tsplib_failure_replay_compression	failure	novelty_gate	True	0.135725	0.0	0.08637	0.757872	0.76	0.009792

Condition Notes

tsplib_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.

- Accepted-epoch count 3, mean accepted code novelty 0.778745, and final complexity 0.56.
- Adaptation efficiency -0.031981 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

tsplib_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.822825, and final complexity 0.56.
- Adaptation efficiency -0.030268 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

tsplib_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.087678, synthetic 0.010343, combined 0.059556.
- Accepted-epoch count 3, mean accepted code novelty 0.80747, and final complexity 0.76.
- Adaptation efficiency 0.058506 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.087678 across 7 instances; family means: ch=0.069489, kroD=0.070583, pcb=0.203691, pr=0.089757, rd=0.036662, st=0.074074.
- Panel synthetic_holdout mean gap 0.010343 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.041372, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3379, "family": "a", "name": "a280", "optimality_gap": 0.310198}, {"best_known_cost": 7542, "cost": 9363, "family": "berlin", "name": "berlin52", "optimality_gap": 0.241448}, {"best_known_cost": 629, "cost": 760, "family": "eil", "name": "eil101", "optimality_gap": 0.208267}]

tsplib_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.135725, synthetic 0.0, combined 0.08637.
- Accepted-epoch count 3, mean accepted code novelty 0.757872, and final complexity 0.76.

- Adaptation efficiency 0.009792 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.135725 across 7 instances; family means: ch=0.102957, kroD=0.132573, pcb=0.174682, pr=0.246406, rd=0.147535, st=0.042963.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3541, "family": "a", "name": "a280", "optimality_gap": 0.373013}, {"best_known_cost": 14379, "cost": 18335, "family": "lin", "name": "lin105", "optimality_gap": 0.275123}, {"best_known_cost": 629, "cost": 778, "family": "eil", "name": "eil101", "optimality_gap": 0.236884}]

Judge Appendix

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Review Summary of Replay-aware TSP Benchmark Suite

Main Evidence: Optimality Gaps and Transfer Metrics

Condition	Mean TSPLIB Gap	Mean Synthetic Gap	Mean Transfer Gap	Replay Mode	Code Novelty (mean)	Completeness	Notes on Transfer and Optimality Gap
tsplib_no_replay	0.213387	0.064916	0.159398	none	0.7787	0.56	Highest TSPLIB gaps and transfer gaps, lowest transfer performance
tsplib_random_replay	0.213387	0.064916	0.159398	random	0.8228	0.56	Identical to no replay in gaps; code novelty decreased slightly
tsplib_failure_replay	**0.087678**	**0.010343**	**0.059556**	failure	0.8075	0.76	Best transfer and TSPLIB gaps; improved optimality and transfer
tsplib_failure_replay_compression	0.135725	0.0	0.08637	failure + compression	0.7579	0.76	Best synthetic gap (0), intermediate TSPLIB and transfer gaps; lower code novelty

Interpretation

Transfer and Optimality

- **Failure replay (tsplib_failure_replay)** achieves the best transfer results, with TSPLIB gap reduced ~59% compared to no replay, and synthetic gap near zero (0.01).
- The **compression-aware variant** of failure replay (tsplib_failure_replay_compression) further reduces synthetic gap to zero but increases TSPLIB and transfer gaps compared to failure replay without compression.
- Both failure replay conditions exhibit notably better optimality gaps on heldout TSPLIB instances than no replay or random replay, indicating that replaying failure cases selectively improves transfer quality.

Code Novelty vs. Transfer

- Code novelty mean slightly drops from no replay (0.7787) and random replay (0.8228) to failure replay compression (0.7579), but transfer metrics improve markedly.
- This suggests that **improved transfer is achieved despite, or even alongside, a decrease in code novelty**.
- Lexical code novelty does not correspond directly to performance improvements; algorithmic gains are supported by deterministic optimality and transfer metrics rather than novelty scores alone.

Replay Mode Effects

- No replay and random replay yield the same (worst) gaps, showing that **random replay does not improve transfer compared to no replay**.
- Selective failure replay (with or without compression) enhances transfer and reduces optimality gaps significantly.
- Compression-aware failure replay slightly compromises transfer compared to failure replay without compression but improves synthetic optimality gap to zero.

Additional Observations

- Complexity increases from 0.56 (no and random replay) to 0.76 (failure replay variants), which might be correlated with improved solution quality.
- Adaptation efficiency is positive only in failure replay conditions, further supporting their effectiveness.
- Worst-case TSPLIB gaps decrease substantially in failure replay: e.g., pr76 gap drops from 0.35 (no replay) to ~0.09.

Summary Conclusion

- **Failure replay conditions provide the strongest evidence for improved transfer and optimality gaps on held-out TSPLIB and synthetic instances.**
- **Random replay does not improve transfer over no replay, despite minor changes in code novelty.**
- Use of compression in failure replay reduces code novelty and increases TSPLIB and transfer gaps slightly compared to failure replay without compression, indicating a trade-off between synthetic optimality and transfer performance.
- Improvements in transfer are supported by deterministic metrics rather than lexical code novelty, confirming that algorithmic advantage stems from selective replay of failure cases rather than mere novelty.